# P1-05 — Formal Proof Package for T5–T6 and Corollaries C1–C4

**Command:** P1-05 — Formal Proof Package for T5–T6 and Corollaries C1–C4 **Date:** 21 October 2026 **Proof source:** P1_04_PROOFS_T1_T4.md (RR-1…RR-5, T1–T4, single component), whose conventions are rechecked at every point of use in §1. **Frozen statements:** transcribed verbatim-in-substance from 02_THEOREM_BLUEPRINT.md before each proof; that file is **not** edited. PC-1…PC-6 remain **unapproved** (register GAP-17).

**Scope.** Labeled exponent assignments only; compatible twists only. This package proves T5, T6 (with the CRT/product assembly made explicit), C1, C2, C3, C4, and the P = 1 specialization of these results. C5, C6, C7 are **not** proved here (see §8); in particular nothing in this file repairs or uses the frozen C5 hull mean, which is known to fail for L_O = 1 (PC-1, unapproved).

---

## 0. Global standing hypotheses

Components are indexed s = 1, …, N with **fixed labels**. Throughout:

- **(G1)** p prime, q = p^e, e ≥ 1, and a **common** Galois index k with 0 ≤ k < e.
- **(G2)** A **common** ϱ ≥ 0, P = p^ϱ, n_0 with gcd(n_0, p) = 1, and common length n = P·n_0.
- **(G3)** K_s = F_{q^{m_s}} = F_{p^{d_s}}, m_s ≥ 1, d_s = e·m_s; A ≅ ∏_{s=1}^N K_s with the isomorphism fixed. A code over A is an A-submodule of A^n; counts are over labeled assignments, never equivalence classes.
- **(G4)** Per component: σ_s(α) = α^{p^k}, ρ_s(α) = α^{p^{d_s−k}} = σ_s^{−1} on K_s (note d_s − k ≥ e − k ≥ 1), λ_s ∈ K_s∗ **compatible**: λ_s^{1 + p^{d_s−k}} = 1; μ_s the unique P-th root of λ_s; S_s = x^{n_0} − μ_s = ∏_{f∈F_s} f (distinct monic irreducibles, (G2)); M_s = S_s^P = x^n − λ_s; τ_s(f) = f# with # as in P1-04 (S6).
- **(G5) Dual-order convention (verbatim, as in P1-04):** “The code-first pairing is ⟨x, y⟩_k = Σ_{i=0}^{n−1} x_i σ(y_i), and the code-first dual is D(C) = { y ∈ K^n : ⟨c, y⟩_k = 0 for all c ∈ C } — the code vector occupies the first slot and the candidate vector the second slot. The Euclidean dual is C^⊥ = { z ∈ K^n : Σ_i c_i z_i = 0 for all c ∈ C }, and Hull_k(C) = C ∩ D(C).” The global pairing on A^n is the tuple of component pairings; D_A and Hull_{A,k} are defined componentwise (justified in §3).
- **(G6)** For a τ_s-orbit O = (f_0, …, f_{L−1}) (with τ_s(f_i) = f_{i+1}, indices mod L_O := L), all factors have the same degree (P1-04, Lemma RR-2(vi)); the weight is w_O = m_s · deg_{K_s}(f_O), and for a factor f, w_f = m_s · deg_{K_s} f.

u, z are indeterminates; all enumerator identities are in Z[u, z].

---

## 1. Inputs from P1-04, with the convention recheck required by this command

The following are **previously proved** (P1-04) and are used as black boxes; the right-hand column records the independent convention check performed for this package.

| Input | Used for | Convention check |
|---|---|---|
| T1: a ↦ ⟨g_a⟩ bijection, dim_K ⟨g⟩ = n − deg g | assignment parametrization in T6 | No dual convention involved. |
| RR-2 / T2: # multiplicative; τ_s permutes F_s, degree-preserving | dual exponents β_i; constancy of w_O on orbits | # uses coefficientwise ρ_s **and** reversal, normalized at the constant term: f# = ρ̃(f^↯_ordinary); it is not the ordinary reciprocal. Multiplicativity is what licenses applying # factorwise to h_a = ∏ f^{P−a_f}. |
| RR-3: C^⊥ = ⟨h^↯⟩ for C = ⟨g⟩, h = (x^n−λ)/g | inside T3/T4, hence inside T5–T6 only through T4 | h^↯ is the **ordinary** reciprocal (no Frobenius), and RR-3 used no separability. The Frobenius enters only afterwards, via RR-4 — keeping ↯ (RR-3) and # (T3) distinct is essential; confusing them would replace β_i by ordinary-reciprocal exponents. |
| RR-4 / T3: D(C(a)) = ⟨h_a#⟩, exponent of f_i equal to β_i = P − α_{i−1} | T4, hence the hull edge weights in T5 | The derivation D(C) = ρ_s(C^⊥) solves Σ c_i σ_s(y_i) = 0 for the **second-slot** candidate y, applying σ_s^{-1} = ρ_s. If the pairing were read candidate-first, the same computation would give σ_s(C^⊥), the σ_s-reciprocal, and β_i = P − α_{i+1}; the frozen h_a# is correct precisely under (G5). Compatibility is used only to keep the dual in the same modulus x^n − λ_s. |

RR-5 / T4: Hull_k(C(a)) = ⟨lcm(g_a, h_a#)⟩; dim_{F_q} C(a) = Σ_s Σ_O w_O Σ_i (P − α_i); dim_{F_q} Hull_k(C(a)) = Σ_s Σ_O w_O Σ_i min(P − α_i, α_{i−1}) | everything below | The orbit orientation τ_s(f_i) = f_{i+1} makes α_{i−1} the **predecessor**; the edge term min(P − α_i, α_{i−1}) pairs current index i (first argument) with predecessor i−1 (second argument). T5's matrix convention (row = previous, column = current) is checked against exactly this in §2.

No hypothesis beyond (G1)–(G6) is introduced anywhere below.

---

## 2. T5 — The transfer-matrix trace identity (RR-6)

**Frozen statement (transcribed).** ∗For one orbit (L, w) let T = T^{(P)}_w(u,z) be the (P+1)×(P+1) matrix with T_{a,b} = u^{w(P−b)} z^{w·min(P−b, a)} (order: previous state a → new state b). Then∗

```
tr(T^L) = Σ_{α∈{0,…,P}^L} u^{w Σ_i (P−α_i)} z^{w Σ_i min(P−α_i, α_{i−1})}.
```

**Theorem T5.** Under (G1)–(G6), for every τ_s-orbit O with L = L_O ≥ 1 and weight w = w_O, the displayed identity holds in Z[u, z], where on the right α = (α_0, …, α_{L−1}), indices are mod L, and the two sums are the per-orbit contributions to dim_{F_q} C(a) and dim_{F_q} Hull_k(C(a)) in T4.

**Proof.** Work in the commutative ring R = Z[u, z]; T has entries in R.

Step 1: trace = closed walks. By the definition of matrix multiplication, induction on L gives, for states a_0, a_L ∈ {0, …, P},

```
(T^L)_{a_0, a_L} = Σ_{a_1, …, a_{L−1}} ∏_{j=1}^{L} T_{a_{j−1}, a_j},
```

(for L = 1 the sum over intermediate states is empty and the formula reads (T)_{a_0,a_1}). Setting a_L = a_0 and summing over a_0 for the trace:

```
tr(T^L) = Σ_{a_0, …, a_{L−1} ∈ {0,…,P}} ∏_{j=1}^{L} T_{a_{j−1}, a_j},   a_L := a_0.     (1)
```

A summand is a **closed walk** a_0 → a_1 → … → a_{L−1} → a_0 through the labeled positions 0, …, L−1 of the orbit. The correspondence with exponent assignments is a bijection: given the walk, set α_i := a_i (i = 0, …, L−1); given α ∈ {0,…,P}^L, the same formula defines a walk, and the closing edge is a_{L−1} → a_0, i.e. the edge into position 0 whose predecessor is position L−1. There is no quotient by rotation: positions are distinct labeled factors f_0, …, f_{L−1}, so walks with different starting positions are different assignments, exactly as T1's parametrization requires. Hence the right side of (1) is a sum over α ∈ {0,…,P}^L, as claimed.

Step 2: the weight of one walk. By the frozen entry definition, with row a_{j−1} = previous exponent and column a_j = current exponent, T_{a_{j−1}, a_j} = u^{w(P − a_j)} z^{w·min(P − a_j, a_{j−1})}. Therefore

```
∏_{j=1}^{L} T_{a_{j−1}, a_j} = u^{w Σ_{j=1}^{L} (P − a_j)} · z^{w Σ_{j=1}^{L} min(P − a_j, a_{j−1})}.
```

Since a_L = a_0, the multiset {a_1, …, a_L} equals {α_0, …, α_{L−1}}, so Σ_{j=1}^{L} (P − a_j) = Σ_{i=0}^{L−1} (P − α_i). Reindexing the hull sum by i := j (with α_L := α_0 and predecessor α_{j−1} = α_{i−1} mod L) gives Σ_i min(P − α_i, α_{i−1}). Substituting into (1) yields the stated identity.

Step 3: consistency with T4. By T4 (§1), the orbit O contributes w Σ_i (P − α_i) to dim_{F_q} C(a) and w Σ_i min(P − α_i, α_{i−1}) to dim_{F_q} Hull_k(C(a)). These are exactly the two exponents above, with the same orientation: current state α_i appears in P − α_i, predecessor α_{i−1} in the second argument of min. (Using the transposed matrix would reverse every walk; the trace is unchanged, but the bijection with assignments would then pair min(P − α_i, α_{i+1}) — a different labeling of the same sum. The frozen row/column convention is the one matching T3–T4, and it is the one used here.)

Case L = 1. There is one position and one (closing) edge, the self-loop: tr(T) = Σ_{a=0}^P T_{a,a} = Σ_a u^{w(P−a)} z^{w·min(P−a, a)}, matching the formula with α_{−1} = α_0. ∎

**Remark.** The proof uses only the definition of matrix multiplication in a commutative ring and T4's local contributions. In particular it is valid for every P ≥ 1, every L ≥ 1, and every w ≥ 1, with no field computation. This discharges RR-6.

---

## 3. The CRT/product assembly (the multi-component step of T6)

**Proposition 3.1 (assembly).** Under (G1)–(G6):

(i) The coefficient-tuple isomorphism A[x] ≅ ∏_s K_s[x] induces A[x]/(x^n − λ) ≅ ∏_s K_s[x]/(x^n − λ_s), where λ = (λ_s)_s.

(ii) Ideals of a finite product of rings are exactly products of ideals of the factors. Hence λ-constacyclic A-codes (ideals of the ring in (i), i.e. A-submodules of A^n closed under the componentwise λ-constacyclic shift) correspond bijectively to tuples (C_s)_s of λ_s-constacyclic codes C_s ⊆ K_s[x]/(x^n − λ_s), and exponent assignments correspond to tuples of component assignments; the total number is ∏_s (P+1)^{|F_s|} = (P+1)^{Σ_s |F_s|} (with T1).

(iii) D_A(C) = ∏_s D_s(C_s) and Hull_{A,k}(C) = ∏_s Hull_k(C_s).

(iv) dim_{F_q} C = Σ_s m_s·dim_{K_s} C_s and dim_{F_q} Hull_{A,k}(C) = Σ_s m_s·dim_{K_s} Hull_k(C_s).

**Proof.** (i) A polynomial over A = ∏_s K_s is a tuple of polynomials over the K_s (coefficients are tuples), giving the ring isomorphism A[x] ≅ ∏_s K_s[x]; the element x^n − λ maps to the tuple (x^n − λ_s)_s, and quotienting a product ring by a product ideal gives the product of quotients. **No coprimality between components is needed**: this is the direct-product decomposition of the square-free algebra A itself, not a Chinese-remainder gluing of coprime moduli inside one component. Inside a component, the only coprimality used is (G2)'s gcd(n_0, p) = 1, which is what makes S_s separable with distinct factors (P1-04). All components share the same length n (G2); without a common n, A^n and a global twist λ would not be defined.

(ii) If I is an ideal of ∏_s R_s, its projections I_s are ideals and I ⊆ ∏_s I_s; conversely, if (x_s)_s ∈ ∏_s I_s, then multiplying by the idempotent e_t (the element of A equal to 1 in component t and 0 elsewhere) shows each (0,...,x_t,...,0) ∈ I, and summing over t gives (x_s)_s ∈ I. Hence I = ∏_s I_s. The correspondence for codes and assignments follows componentwise from T1 (P1-04), and the count multiplies.

(iii) By (G5) the global pairing is componentwise: for x = (x_s), y = (y_s), the pairing value is the tuple (⟨x_s, y_s⟩_{s,k})_s ∈ A, which is 0 in A iff every component is 0. Hence y ∈ D_A(C) iff y_s ∈ D_s(C_s) for all s (using (ii) to see c ranges over C iff each c_s ranges over C_s). Intersecting with C = ∏_s C_s componentwise gives the hull statement. Compatibility (G4) is what places each D_s(C_s) inside the same component quotient K_s[x]/(x^n − λ_s) (P1-04, T3); without it, (iii) still defines a product of transformed-twist codes, but the T6 formula in a single modulus is not asserted.

(iv) As F_q-vector spaces, A^n ≅ ⊕_s K_s^n (direct sum of the components), a product subspace has dimension equal to the sum of the component dimensions, and dim_{F_q} V_s = [K_s : F_q]·dim_{K_s} V_s = m_s·dim_{K_s} V_s for any K_s-subspace V_s. Apply this to C and to Hull_{A,k}(C) via (iii). ∎

---

## 4. T6 — The global joint enumerator (RR-7)

**Frozen statement (transcribed).** Under the hypotheses above,

```
E(u,z) := Σ_{labeled a} u^{dim_{F_q} C(a)} z^{dim_{F_q} Hull_k(C(a))}
        = ∏_{s=1}^N ∏_{O∈O_s} tr( T^{(P)}_{w_O}(u,z)^{L_O} ).
```

**Theorem T6.** Under (G1)–(G6), the displayed identity holds in Z[u, z].

**Proof.** By Proposition 3.1(ii) and T1, the set of labeled global assignments is the Cartesian product Aset = ∏_{s=1}^N ∏_{f∈F_s} {0, ..., P} (the set of labeled global assignments), and every a ∈ Aset is uniquely the disjoint union of its orbit restrictions α^{(O)} ∈ {0,...,P}^{L_O} over all pairs (s, O). By T4 (with Proposition 3.1(iii)–(iv) converting component dimensions to F_q-dimensions),

```
dim_{F_q} C(a)      = Σ_{s} Σ_{O∈O_s} w_O Σ_{i=0}^{L_O−1} (P − α^{(O)}_i),
dim_{F_q} Hull_k(C(a)) = Σ_{s} Σ_{O∈O_s} w_O Σ_{i=0}^{L_O−1} min(P − α^{(O)}_i, α^{(O)}_{i−1}),
```

i.e. each global dimension is a **sum over (s, O) of a quantity depending only on the orbit's**

**own exponent tuple**. For each orbit define the local polynomial (in Z[u, z]) E_O(u, z) := Σ_{α∈{0,...,P}^{L_O}} u^{w_O Σ_i (P−α_i)} z^{w_O Σ_i min(P−α_i, α_{i−1})}. By T5, E_O(u, z) = tr( (T^{(P)}_{w_O}(u,z))^{L_O} ).

It remains to factor the global sum. This is the finite distributive law: if an index set is a finite Cartesian product ∏_j J_j and the exponent of each indeterminate is a sum Σ_j Φ_j(α^{(j)}) over the factors, then Σ_{α ∈ ∏_j J_j} ∏_j u^{Φ_j(α^{(j)})} = ∏_j ( Σ_{α^{(j)} ∈ J_j} u^{Φ_j(α^{(j)})} ) (both sides expand to the same finite sum: the left side is the product of the right-side sums multiplied out; for two factors this is Σ_{(x,y)} φ(x)ψ(y) = (Σ_x φ(x))(Σ_y ψ(y)), and the general case follows by induction on the number of factors). Applying this with j = (s, O), J_j = {0,...,P}^{L_O}, and the two exponent functions above (one for u, one for z, weights w_O included):

E(u, z) = ∏_{s=1}^N ∏_{O∈O_s} E_O(u, z) = ∏_{s=1}^N ∏_{O∈O_s} tr( (T^{(P)}_{w_O}(u,z))^{L_O} ).  ∎

**Hypotheses actually consumed (RR-7 checklist).** Common P, n, k and per-component compatibility (G1, G2, G4) — compatibility only via T3/T4; common length n for the product A^n (G2); fixed labeled decomposition with A-submodule codes (G3, Proposition 3.1(ii)); gcd(n_0, p) = 1 only through the separable base S_s (G2/G4). This discharges RR-7.

---

## 5. Corollaries C1–C4

Each corollary is first transcribed from the frozen blueprint, then derived.

### C1 — Total count

**Frozen statement.** E(1,1) = (P+1)^{Σ_s |F_s|}.

**Derivation.** At u = z = 1, every entry of T = T^{(P)}_{w_O}(1,1) equals 1, i.e. T = J, the (P+1)×(P+1) all-ones matrix. J^2 = (P+1)J (each entry of J^2 is a sum of P+1 ones), so J^{L} = (P+1)^{L−1} J for L ≥ 1 by induction, and tr(J^L) = (P+1)^{L−1}·(P+1) = (P+1)^L. (Equivalently, by T5, tr(T^L) at u=z=1 counts the (P+1)^L tuples α.) By T6, E(1,1) = ∏_s ∏_O (P+1)^{L_O} = (P+1)^{Σ_s Σ_O L_O} = (P+1)^{Σ_s |F_s|}, since the orbits of s partition F_s, so Σ_O L_O = |F_s|. This agrees with the direct count in Proposition 3.1(ii). ∎

### C2 — Code-dimension distribution

**Frozen statement.** E(u,1) = ∏_s ∏_{f∈F_s} (1 + u^{w_f} + ⋯ + u^{P w_f}), where w_f = m_s·deg f.

**Derivation.** Set z = 1 in T6. In T = T^{(P)}_w(u, 1), the entry T_{a,b} = u^{w(P−b)} does not depend on the row a. Write c = (1, ..., 1)^T (column) and r for the row vector with r_b = u^{w(P−b)}; then T = c·r (outer product). Hence T^L = (r·c)^{L−1}·T (each internal contraction contributes the scalar r·c), and tr(T^L) = (r·c)^{L−1}·tr(T). Now r·c = Σ_{b=0}^P u^{w(P−b)} = Σ_{j=0}^P u^{w j} (substituting j = P − b), and tr(T) = Σ_{a=0}^P u^{w(P−a)} = Σ_{j=0}^P u^{w j} as well. Therefore tr( (T^{(P)}_w(u,1))^L ) = ( Σ_{j=0}^P u^{w j} )^L. For an orbit O, every factor has the same degree (RR-2(vi)), hence the same weight w = w_f for all f ∈ O, and the orbit contributes (Σ_{j=0}^P u^{w_f j})^{L_O} = ∏_{f∈O} (1 + u^{w_f} + ⋯ + u^{P w_f}). Taking the product over all orbits and components (T6) gives exactly ∏_s ∏_{f∈F_s} (1 + u^{w_f} + ⋯ + u^{P w_f}). ∎

(Consistency: at u = 1 this returns C1.)

### C3 — LCD count

**Frozen statement.** #{a : Hull = 0} = 2^{#orbits}. (Here Hull = 0 means dim_{F_q} Hull_{A,k}(C(a)) = 0, and #orbits is the total number of τ_s-orbits over all components.)

**Derivation.** dim_{F_q} Hull_{A,k}(C(a)) = 0 iff every orbit contributes 0 in T4's formula: the total is a sum of terms w_O·min(P − α_i, α_{i−1}) with w_O ≥ 1 and min(·,·) ≥ 0, so the sum vanishes iff min(P − α_i, α_{i−1}) = 0 for every orbit and every i (equivalently, this is the coefficient of z^0 in E(1, z)).

Fix one orbit and suppose $\min(P - \alpha_i, \alpha_{i-1}) = 0$ for all $i$. **Claim: $\alpha$ is either identically 0 or identically P.** Suppose some $\alpha_m \notin \{0, P\}$ (possible only if $P \geq 2$). Since $\min(P - \alpha_{m+1}, \alpha_m) = 0$ and $\alpha_m > 0$, we must have $P - \alpha_{m+1} = 0$, i.e. $\alpha_{m+1} = P$. Then $\min(P - \alpha_{m+2}, \alpha_{m+1}) = \min(P - \alpha_{m+2}, P) = P - \alpha_{m+2} = 0$ forces $\alpha_{m+2} = P$; continuing around the finite cycle, every $\alpha_i = P$, contradicting $\alpha_m \notin \{0, P\}$ (for $L = 1$ the starting supposition already gives $\min(P - \alpha_0, \alpha_0) = 0$ only for $\alpha_0 \in \{0, P\}$ when $P \geq 2$, and for $P = 1$ there are no interior values at all). Hence every $\alpha_i \in \{0, P\}$. If both values occur, choose $i$ with $\alpha_i = 0$ whose predecessor has $\alpha_{i-1} = P$ (such a boundary index exists on a cycle containing both values); then $\min(P - \alpha_i, \alpha_{i-1}) = \min(P, P) = P \geq 1$, contradiction. So $\alpha \equiv 0$ or $\alpha \equiv P$: exactly **2** admissible tuples per orbit, for every $P \geq 1$ — the count is independent of P. Choices on distinct orbits/components are independent (T1, Proposition 3.1(ii)), so the total count is $2^{\#\text{orbits}}$. ∎

(This proof replaces the frozen one-line sketch, whose forward-propagation wording is imprecise for interior exponents; the **statement** needed no correction.)

### C4 — Self-orthogonal and dual-containing characterizations

**Frozen statement.** $C \subseteq D(C) \iff$ on every orbit edge $\alpha_{i-1} + \alpha_i \geq P$; the count is $\prod_O \mathrm{tr}(B_P^{L_O})$ with $(B_P)_{a,b} = 1$ if $a + b \geq P$, else 0. Conversely $D(C) \subseteq C \iff \alpha_{i-1} + \alpha_i \leq P$ on every edge. (The frozen text states no count for the dual-containing side; see PC-4 below.)

**Derivation.** Work globally; all dimension identities are over $\mathbb{F}_q$ via T4/Proposition 3.1.

Self-orthogonal side. By definition $\mathrm{Hull}_{A,k}(C) = C \cap D_A(C) \subseteq C$, so $C \subseteq D_A(C)$ iff $C \cap D_A(C) = C$ iff $\dim \mathrm{Hull} = \dim C$ (a subspace equals a containing subspace iff dimensions agree). By T4,

```
dim C − dim Hull = Σ_{s,O} w_O Σ_i [ (P − α_i) − min(P − α_i, α_{i−1}) ].
```

For each edge, $(P - \alpha_i) - \min(P - \alpha_i, \alpha_{i-1}) = \max(0, P - \alpha_i - \alpha_{i-1})$: if $\alpha_i + \alpha_{i-1} \leq P$, then $\min(P - \alpha_i, \alpha_{i-1}) = \alpha_{i-1}$ and the difference is $P - \alpha_i - \alpha_{i-1} \geq 0$; if $\alpha_i + \alpha_{i-1} \geq P$, then $\min(P - \alpha_i, \alpha_{i-1}) = P - \alpha_i$ and the difference is 0. Each summand is $\geq 0$ and $w_O \geq 1$, so the total is 0 iff every edge satisfies $\alpha_{i-1} + \alpha_i \geq P$. This proves the characterization.

For the count: the condition factorizes over orbits. On one orbit, admissible tuples are the closed walks on states $\{0,\ldots,P\}$ whose every step $a \to b$ satisfies $a + b \geq P$; by the trace expansion of §2 Step 1 applied to the 0/1-matrix $B_P$ (same computation with weights 0/1), their number is $\mathrm{tr}(B_P^{L_O})$. Independence over orbits (T1) gives the global count $\prod_O \mathrm{tr}(B_P^{L_O})$ (product over all components' orbits).

Dual-containing side. Again $\mathrm{Hull} = C \cap D_A(C) \subseteq D_A(C)$, so $D_A(C) \subseteq C$ iff $\dim \mathrm{Hull} = \dim D_A(C)$. By T3, on one orbit the dual generator has exponents $\beta_i = P - \alpha_{i-1}$, so T4 applied to the dual assignment gives $\dim D(C(a)) = \Sigma_{s,O} w_O \Sigma_i (P - \beta_i) = \Sigma_{s,O} w_O \Sigma_i \alpha_{i-1} = \Sigma_{s,O} w_O \Sigma_i \alpha_i$ (reindexing $i \mapsto i+1$ in the last sum; the orbit sum is cyclic). Hence

```
dim D(C) − dim Hull = Σ_{s,O} w_O Σ_i [ α_{i−1} − min(P − α_i, α_{i−1}) ],
```

and for each edge, $\alpha_{i-1} - \min(P - \alpha_i, \alpha_{i-1}) = \max(0, \alpha_i + \alpha_{i-1} - P)$: if $\alpha_i + \alpha_{i-1} \leq P$, then $\min(P - \alpha_i, \alpha_{i-1}) = \alpha_{i-1}$ and the difference is 0; if $\alpha_i + \alpha_{i-1} \geq P$, then $\min(P - \alpha_i, \alpha_{i-1}) = P - \alpha_i$ and the difference is $\alpha_i + \alpha_{i-1} - P \geq 0$. So $\dim D(C) - \dim \mathrm{Hull} = 0$ iff every edge satisfies $\alpha_{i-1} + \alpha_i \leq P$, proving the converse characterization. ∎

**Proposed completion PC-4 (unapproved; not part of frozen C4).** The same closed-walk argument gives $\#\{a : D(C) \subseteq C\} = \prod_O \mathrm{tr}((B'_P)^{L_O})$ with $(B'_P)_{a,b} = 1$ iff $a + b \leq P$. Moreover the two counts in C4 are equal: the involution $\alpha_i \mapsto P - \alpha_i$ on each orbit maps $\{\alpha_{i-1} + \alpha_i \geq P\}$ bijectively to $\{\alpha_{i-1} + \alpha_i \leq P\}$ (as $(P - \alpha_{i-1}) + (P - \alpha_i) \leq P \iff \alpha_{i-1} + \alpha_i \geq P$), preserving closed walks. Adding the displayed count to the frozen corollary remains subject to the PC approval command; the **characterization** above is frozen and is proved as stated.

---

## 6. Simple-root recovery: the P = 1 specialization (Gate G3 bookkeeping for T5/T6/C1–C4)

Set ϱ = 0, so P = 1. Then:

- **Model.** μ_s = λ_s (the unique 1st root), M_s = S_s = x^{n_0} − λ_s, which is separable with distinct factors (gcd(n_0, p) = 1); assignments are binary, a_f ∈ {0, 1}, and g_a is a squarefree divisor — the classical simple-root constacyclic parametrization. σ_s, ρ_s, τ_s are unchanged (they depend on d_s, k, not on P), and compatibility is the same condition.
- **Local terms (T4 at P = 1).** P − α_i = 1 − α_i, and min(1 − α_i, α_{i−1}) equals 1 if (α_{i−1}, α_i) = (1, 0) and 0 otherwise: the hull contribution of an edge is the indicator of the transition 1 → 0. Hence dim Hull counts, with weights w_O, the number of 1 → 0 transitions around each orbit — the predecessor (simple-root) hull formula in this notation.
- **Transfer matrix (T5 at P = 1).** With states {0, 1} (row = previous a, column = current b), T^{(1)}_w(u,z)_{a,b} = u^{w(1−b)} z^{w·min(1−b, a)} gives

```
T^{(1)}_w(u,z) = ⎡ u^w        1   ⎤      (a=0 row)
                 ⎣ u^w z^w    1   ⎦      (a=1 row)
```

since min(1,0) = 0, min(0,0) = 0, min(1,1) = 1, min(0,1) = 0. The T6 product ∏_s ∏_O tr( (T^{(1)}_{w_O})^{L_O} ) is therefore exactly the predecessor binary-state joint enumerator, with the same orbit/orientation conventions (τ(f_i) = f_{i+1}, predecessor α_{i−1}, closing edge from position L−1 to position 0). This proves algebraically — not just by analogy — that the P = 1 specialization of T5/T6 **is** the stated simple-root formula.

- **Corollaries at P = 1.** C1: E(1,1) = 2^{Σ_s |F_s|} (all squarefree divisors). C2: E(u,1) = ∏_s ∏_f (1 + u^{w_f}). C3: 2^{#orbits} (unchanged — C3 is P-independent). C4: self-orthogonal iff no edge is (0,0) (i.e. α_{i−1} + α_i ≥ 1 on every edge), dual-containing iff no edge is (1,1) (i.e. α_{i−1} + α_i ≤ 1 on every edge), counted by tr(B_1^{L_O}) resp. ∏ tr((B′_1)^{L_O}) (the latter via PC-4).

**Scope note.** This recovery concerns T5, T6, C1–C4 only. It deliberately makes no claim about the frozen C5 remark "P = 1 gives 1/4": that remark is false for L_O = 1 orbits (ν_1 = 0), is already recorded (GAP-01/PC-1), and is not used anywhere above.

---

## 7. Dependency map

**Previously proved (P1-04; used as inputs, conventions rechecked in §1):**

```
Fact 0.1/0.2 (K[x] PID/UFD) ─► RR-1 ─► T1 ──────────────┐
RR-2 (#, τ) ─► T2 ─► T3 (with RR-3, RR-4) ─► T4 ────────┤
RR-3 (Euclidean dual) ─► T3          RR-5 (lcm) ─► T4 ──┘
```

**New in this package:**

```
T4 (orbit-local dimension sums) ─► T5 [trace = closed walks; RR-6 discharged] ─┐
Prop. 3.1 (CRT/product assembly; ideals, dual, dimensions) ──────────────────┤
Finite distributive law ──────────────────────────────────────────────────────┴► T6 [RR-7 discharged]
T6 + T5 (J-power / rank-one trace) ─► C1, C2
T4 (hull sum = 0 termwise) ─► C3
T4 + T3 (dim D via β_i; nonnegative edge differences) ─► C4
§2–§5 at ϱ = 0 ─► P = 1 recovery (2×2 matrix; 1→0 transition indicator)
```

**Distinguishing old from new:** T5's only new content is the closed-walk expansion (RR-6); T6's only new content is Proposition 3.1 plus the distributive law (RR-7). Both are proved above in full; neither relies on any computation.

---

## 8. Remaining obligations (not proved here; no claim made)

1. **C5** — frozen hull mean is incorrect for L_O = 1; the corrected split (μ_P for L_O ≥ 2, ν_P for L_O = 1) is candidate **PC-1**, unapproved. The F_q code-dimension mean in frozen C5 is correct but is not re-proved here (P1-03 §4.2 records the derivation).
2. **C6 / RR-8** — variance closed forms (L = 1, L = 2, L ≥ 3 covariances) remain open; exact finite-sum obligations are listed in P1-03 §4.3.

3. **C7 / RR-9** — conditional EAQECC statement only; the enumerator alone yields no distance or quantum parameter.

4. **PC-1…PC-6** — all unapproved. PC-4's dual-containing count is proved above as a labeled proposal; PC-2/PC-3 concern blueprint wording (the dual-order sentence; f(0) ≠ 0) adopted inside P1-04/P1-05 hypotheses only.

5. **Gate G1** — literature/novelty remains open and untouched by these proofs; no novelty or priority claim is made.

---

## 9. Computational cross-check (supporting evidence, not proof)

Executed for this package (21 Oct 2026); existing artifacts unmodified:

```
python3 code/validate_p1_03.py            # rerun: exit 0, output byte-identical to
code/validation_p1_03_output.txt
                                  # (single-component T5/T6 + C1-C4 vs Gram-rank brute force;
                                  #  6 vector cases, 2,806 assignments; 31 (P,L) orbit pairs)
python3 code/validate_p1_05.py    # NEW script; output code/validation_p1_05_output.txt
                                  # final line: ALL P1-05 MULTI-COMPONENT ASSEMBLY CHECKS PASSED
```

The new P1-05 check is the first **multi-component** test: A = GF(2) × GF(4) (q = 2, k = 0, λ = (1,1), P = 2, n_0 = 7, n = 14; component weights m_1 = 1, m_2 = 2), 729 labeled global assignments. It verifies that the joint F_q-dimension distribution obtained by adding independently brute-forced (Gram-rank) component dimensions equals (i) the convolution of the component distributions, (ii) the T6 product of per-orbit transfer traces with weights w_O = m_s·deg f_O (orbits (L,w) = (1,1), (2,3) and (1,2), (2,6)), and (iii) the C1 total 3^6 = 729, the C2 marginal, C3's count 2^4 = 16, and C4's self-orthogonal/dual-containing counts 144 = 144. What it cannot prove: the CRT/A-submodule correspondence (Proposition 3.1 is a proof, not a computation), the T5/T6 formulas in general, or anything for k > 0, λ ≠ 1, P ≠ 2, or N > 2. Finite agreement is supporting evidence only.